

SOLVING PRACTICAL VEHICLE ROUTING PROBLEM WITH TIME WINDOWS (VRPTW) – A CASE STUDY OF ICD TIEN SON (VIETNAM)

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Abstract

In logistics, there are a range of methods to deal with vehicle routing problem with the optimization directions related to cost, distance, vehicle and so on. In fact, a good route network plays an important role in serving efficiently demands of customers who have own different time ranges of receiving goods. In the research process, authors applied the vehicle routing problem with time windows (VRPTW) to propose the route network towards minimizing the total travelled distance and meeting constraints of time windows, vehicles (capacity and number), allowed time of operation on urban road. Subsequently, the suggested routes compared with the present network of ICD Tien Son based on operating parameters (travelled distance, load factors, travelled time...) to discover what the better choice is? With the support of C++ software, seven routes serving 25 positions were constructed. In comparison with the ICD Tien Son's network, the number of routes and trucks decrease by one and one respectively, the total travelled distance presents an about 8% reduction however the total time of completing transport duties is the same because of a rise in the waiting time (approximately 2.5%) at customers' points. For the above analysis, the result of VRPTW application is better because it carries all freight in the same time with fewer vehicles, drivers and a smaller number of travelled kilometers.

Keywords: VRPTW; Vehicle routing problem; Routes; Travelled distance; ICD Tien Son

1. Introduction

ICD Tien Son is located in a strategic location – at the heart of the Northern Delta and only 21 Km far from Hanoi Capital. What is more, it is on the crossroads of all the main highways and railways leading to major seaports, China and the South of Vietnam. It was developed under the concept of “Inland Logistics Supermarket” with the highest quality infrastructure in Vietnam and the fullest scope of services including warehousing, transportation, customs, container yard and ancillary services, which cater the demanding needs of different customers.



Fig. 1 The strategic location of ICD Tien Son

As of 2013, the total area of ICD Tien Son is 12 ha, consisting of 45,000 m² of European A-class warehouse, 40,000 m² of container yard with 4,000 TEU capacity, 2,000 m² of office space, a mechanic 's repair workshop for trucks, containers and a designed parking area. By 2015, the vehicle fleet also rose to 85 trucks (included heavy and light ones) and 120 trailers. It also built 20,000 m² of new warehouse to increase total warehouse up to 65,000 m². All warehouse system has been well-equipped of modern flat floor with 9 ton /m² of capacity, 6 floor shelves, forklifts that can reach 20 meters and 120 tons of electronic weighing stations. Thanks to favorable geographical conditions and big investment in facilities, ICD Tien Son has cooperated with many clients and partners in the same fields: Nissin Logistics Co.; Seeds CP-VN; Yusen Logistics Solution, Panapilna World Transport, K Logistics VN-line... Also included in the list are big customers such as: Samsung, Canon, Ikea...

Table 1 The annual financial records (*Unit: USD*)

	Records	2013	2014	2015
1	Turnover from transportation service	418,182	586,909	1,328,000
2	Turnover from warehousing service	976,727	1,372,455	1,422,545
3	Turnover from container yard service	104,091	134,727	184,909
4	Turnover from customs service	46,000	71,000	52,364
5	Turnover from different client services	54,091	77,455	70,000
6	Total turnover	1,612,727	2,283,636	3,057,818
7	Cost	1,423,545	2,025,455	2,417,455

Source: Annual reports of ICD Tien Son

Along with services related to container, distribution of packages from the warehouse in Bac Ninh to customers in urban areas, especially the center of Hanoi capital is more and more important. The turnover from storing and transporting these packages increased constantly to about 8% of the total in 2015.

ICD Tien Son is facing difficulty in constructing routes to meet clients' needs. In general, the problem is complicated because every customer has own requirements (unloading time, number of packages). It wants to join positions (customers) into the small number of routes to save cost related to vehicles, drivers and so on. However, how to group clients effectively is challenging. The

present way to construct routes is quite simple. Dispatchers usually based on their experience to propose the routes with the priority for connecting between the furthest points and depot. Undeniably, this method is still particularly useful however there is no evidence to demonstrate it is ideal. Currently, there are mathematical methods to form routes in logistics. Therefore it is necessary to apply one popular mathematical method to seek a new solution which is likely better than the current practice.

2. Literature Review

2.1. Vehicle Routing Problem with Time Windows (VRPTW)

The vehicle routing problem (VRP) first appeared in a paper by George Dantzig and John Ramser in 1959. In VRP, the objective is to design a set of m vehicle routes that minimizes the total traveled distances.

The vehicle routing problem with time windows (VRPTW) is an extension of the well-known vehicle routing problem (Crainic and Laporte, 2000), (Toth and Vigo, 2002). The VRPTW is an important generalization of the VRP and a basic distribution management problem that can model many real-world problems which consists of designing a set of minimum cost routes, originating and terminating at a central depot, for a fleet of vehicles which serves a set of customers with known demands. The customers must be assigned exactly once to vehicles such that the vehicle capacities are not exceeded. The service at a customer must begin within the time window defined by earliest time and the latest time when the customer permits the start of service. Some of the more useful applications of the VRPTW include bank deliveries, postal deliveries, industrial refuse collection, national franchise restaurant deliveries, and school bus routing and security patrol services (Nasser, 2010).

Knowledge-based Locomotive Planning for the Swedish Railway by Volker Scholz (1998). The customers can additionally specify a time window for the delivery time. Due to the time windows, a vehicle can arrive to early at a customer location and has to wait until the earliest delivery time has been reached. These waiting times are additional cost factor.

To deal with VRP, in 1964, Clarke & Wright published an algorithm for the solution of that kind of vehicle routing problem, which is often called the classical vehicle routing problem. This algorithm is based on a so-called savings concept.

The savings algorithm is a heuristic algorithm, and therefore it does not provide an optimal solution to the problem with certainty. The method often yields a relatively good solution. That is, a solution which deviates little from the optimal solution.

For VRPTW, very limited work has been done on optimal approaches to solve it with practical size, while heuristics have been more effective and efficient (Salomon and Desrosiers, 1988). The fact shows complex problems, therefore the algorithm have to be able to solve the problem with two dimensions which are space and time. The original heuristic algorithm (of Clarke and Wright) is only dealing with one dimension that is the space (or distance). To put it another way, to deal with VRPTW, it is necessary to improve the Clarke and Wright's method through adding constraints related to time window at every customer's point.

2.2. Application of VRPTW in Vietnam

VRP in Vietnam is quite popular content in the curriculum at universities researching on road transport and logistics. Therefore, studies on application of VRP in real life have been carried out quite frequently by students and researchers. While, VRPTW is a much complicated version of VRP, hence there have been fewer studies on it. Some typical ones are as follows:

In 2000, Tung and Pinnoi at Asian Institute of Technology addressed the vehicle routing-scheduling problem with the case of waste collection in Hanoi. The problem is complicated because of several time windows and inter-arrival time constraints at each customer point. The route construction is a modification of Solomon's heuristic along with the power of Or-opt and 2-opt*. Compared with current practice, their proposed heuristic shows a good performance in terms of the solution quality and computational time.

In 2005, Mai at University of Transport and Communications composed the thesis with the financial support of Education Ministry to improve the effectiveness of collecting waste in two districts of Hanoi (which is the basic VRP). Gathering and carrying waste are implemented every day by manually pushed handcarts before loaded to motorized trucks. She applied the savings algorithm with the support of Excel to find the solutions which have the less total travelled distance theoretically. However, some opposites claimed on her way to calculate the distance between locations, especially in the one-way road. The demand of vehicles was also a problem because solution was

proposed with the highest priority of choosing trucks.

In the thesis for master degree, Truong (2009) conducted a research on carrying dangerous waste in Ho Chi Minh city. Because this is the special waste, the way to collect it is also more careful and stricter. A truck has to go to factories to collect waste before taking it to a waste treatment plant. Subsequently, it could repeat these activities through bringing waste at following factories to (other) plants. In this sense, one truck can take responsible for more than one route (transport mission). Every factory and plant has its own accepted time range for dumping and transporting, therefore the case is the VRPTW. To resolve it, Truong started with the classical heuristic algorithm of Clarke and Wright before improving the solution through using the neighborhoods of VRPTW and other algorithms (Hill Climbing, Tabu, Simulated Annealing) and the assist of C++ software. The hole of this study is there is no comparison to assess the result of VRPTW.

For the above-analysis, application of VRPTW in Vietnam is dominant in the field of collecting and transporting waste. In logistics, especially distribution packages in urban areas, VRPTW's application is quite limited.

3. Methodology and Data

3.1. Methodology

Transport mission in this research is responsible for by ICD Tien Son in Bac Ninh province with one depot and 25 customers having different geographical positions in Hanoi capital and its own different time range to receive goods. This is the vehicle routing problem with time windows which is usually dealt with through the mathematical procedures. Authors will use the most popular heuristic algorithm of Clarke and Wright to start constructing route network and utilize capacity and time windows constraints to seek the solutions. Last, based on the comparison between the current practice and the result of VRPTW to find what the better is.

Step 1: Collect the distance data: The distances between locations of customers and depot are calculated with the records of company (if the roads/routes are operating) and the support of Google Maps which is a currently popular web mapping service.

Step 2: Calculate distance savings: Distance savings are calculated through the Wright and Clark's mathematical formula as follows

$$S_{ij} = D_{oi} + D_{oj} - D_{ij} \quad (1)$$

$i=1,2,3,\dots,n; j=1,2,3,\dots,n; i \neq j$

Where

S_{ij} : Distance saving between point i and point j

D_{oi} : Distance between depot and customer location i

D_{oj} : Distance between depot and customer location j

D_{ij} : Distance between customer location i and customer location j

All distance savings subsequently are stored in the matrix.

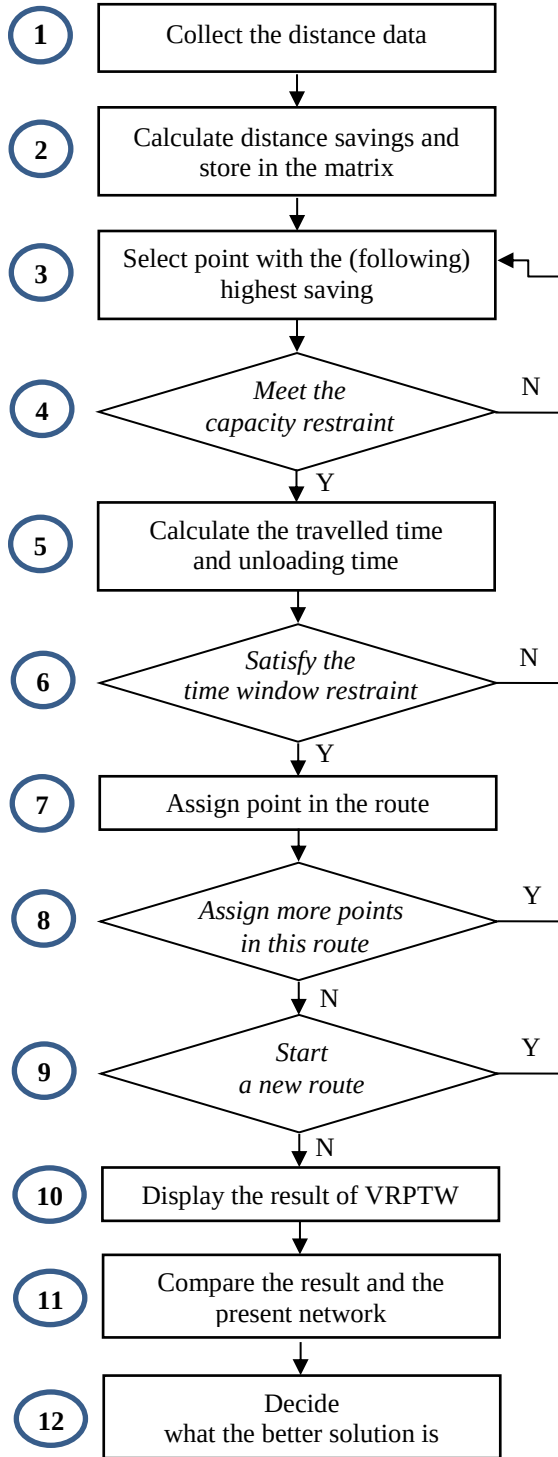


Fig. 2 The research process

Step 3: Select point with the (following) highest saving: With the objective to minimize the total travelled distance, the point with the largest distance saving is prioritized to be chosen. During the process, all the points are experimented in the descending turn of distance savings. In other words, after a point with the highest distance saving is selected or eliminated, the one having the following highest distance saving will be tested.

Step 4: Check the capacity condition: A truck is not allowed to be overloaded, therefore if the customer's demand and the goods on the vehicle are equal or less than truck capacity, this point is kept for the step 5. If the point fails to satisfy the constraint, it will be eliminated and the process returns the 3rd stage.

Step 5: Calculate the travelled time and unloading time: The travelled time is dependable on the distance between points and speed of vehicle. It is noticeable that speed is variable because of the different real conditions of road. For example, a truck can run at higher speed in expressways and highways where the speed restrictions are larger significantly than these in urban road.

$$T_{i-j}^{run} = \sum \frac{D_k}{S_k} \quad (2)$$

Where

T_{i-j}^{run} : Travelled time to run from location i to location j

D_k : The length of the section k (of the distance $i-j$)

S_k : The speed of vehicle on the section k

The unloading time plays an important role in constructing routes because it affects the moment when vehicle possibly reaches the following customer's point.

$$T_i^{Unload} = t_{unit}^{unload} * Q_i \quad (3)$$

Where

T_i^{unload} : Unloading time in the customer location i

Q_i : Demand of customer i

t_{unit}^{unload} : Unloading time of one unit

Step 6: Check the time window constraint: As indicated above, one location has its own possible unloading time, therefore if the moment when the truck reaches the point is not soon enough, which leads the unloading time to exceed the acceptable latest moment and consequently, this point is eliminated. Unloading goods has to finish in the time window. If it does not satisfy this condition, return the step 3.

Step 7: Assign point in the route: When the testing point meets two constraints, it will be on the route. And go to step 8.

Step 8: Check the possibility of adding more point in the route: This step basically is the minor process from step 3 to step 7. After one point

is added in the step 7, come back to step 3 (to find the following point with highest distance saving), subsequently it is experimented through two constraints of capacity and time window. If it overcomes both two tests, it will be a part of the route. By contrast, it will be removed. All the rest of points are checked in the same procedures. If no point can be added, the route is completed. Go to step 9

Step 9: Check the possibility of starting a new route: In the step, except for points on the constructed routes, remaining points will participate from step 3 to step 8 in turn. If no point satisfies to become the first point of a new route, this means that every point is experimented and arranged in route network. Go to step 10.

Step 10: Display the result of VRPTW: All routes with ordered points are showed in this stage.

Step 11: Compare the result and the present network: the most effective way to obtain the solution to VRPTW is an approximately method (or heuristic). In other words, it is not persuasive if the result is considered to be better than other solutions.

The basement of comparison is concept of effectiveness which is evaluated through output and input. In transportation, input possibly is some operating parameters included: the number of trucks, the number of drivers, the number of routes, the total travelled distance, the unloading time, the travelled time, the time to complete the transport mission, load factor, cost... Noticeably, the number of travelled distance, in some cases, is not in proportional to the travelled time due to the difference in speed levels. Output can be the completion of transport missions or serve all the customers' demands in right time, right place and meet all the constraints.

$$\gamma = \frac{\sum \left(\frac{q_i}{C} l_i \right)}{L} \quad (4)$$

Where

- γ : Load factor in the route
- q_i : The temporary load of truck
- C : Capacity of vehicle
- l_i : Length of the section i
- L : Length of the route

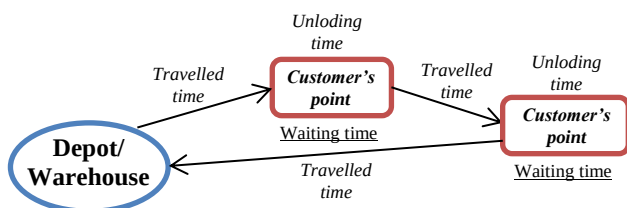


Fig. 3 Describe of transport mission

Transport mission is the process in which vehicle begins carrying freight from the depot to all targeted customers' points before returning the depot. In this case, the time for loading in the depot is not concerned because warehouse and depot are located in the same position. So the goods can be loaded in prior to start running for a long time, in the previous day for example. The time to complete transport mission is the time of aforesaid process.

$$T_{T-M} = \sum (t_{travel} + t_{wait} + t_{unload}) \quad (5)$$

Where

- T_{T-M} : Total time to complete transport mission
- t_{travel} : Time for travelling to customer's point/depot
- t_{wait} : Time for waiting to unload in customer's point
- t_{unload} : Time for unloading at customer's point

Step 12: Demonstrate what the better is:

With same output, which requires less input is the better and the chosen solution.

3.2. Data

In this case, distance data is collected through the support of Google Maps and the statistical figures of ICD Tien Son

As regards the goods, they are cosmetic products of Loreal enterprise. They are arranged in the carton packages to distribute from warehouse of ICD Tien Son to shops, supermarkets in Hanoi. The characteristics of cosmetic packages are that they should be loaded and unloaded softly and primarily use the volume of vehicle boot rather than the quantity.

Customers' locations: There are 25 positions in Hanoi with the demands changing from 2 to 4 packages and 6 – 8 packages for small shops and supermarket respectively. Transport happens every two days.

Capacity and number of vehicle: There is strict restriction on heavy trucks in urban areas during daytime, hence the light trucks with the capacity of 1 ton are chosen because they can run at any time. One truck can load a maximum of 15 packages (because there is no space to arrange more). The maximum number of available vehicles is 8 and one truck only serves one route.

Speed: It is divided into two types based on the areas where vehicles run. The average speed in highway from Bac Ninh province to the suburb of Hanoi capital is 45 Km/h. While it in center of the capital is only 30 Km/h.

Time of unloading goods: This time depends on the number of packages and time of unloading one package at every point. Time of unloading one package is considered to be the same at all points. Besides time for carrying package

from the truck to customer's shop/warehouse, time for counting and checking every cosmetic product in a package also form unloading time. The survey result showed that this time is about 10 - 15 minutes for one package.

Time windows: Customer accepts to receive goods in own time range.

The starting time: There is no restriction on the starting time. Trucks can begin at particular time to unload time at the first point without waiting time.

The support of C⁺⁺

Solving the VRPTW with 25 positions and 1 depot is a heavy task, therefore authors utilized the C⁺⁺ software to find the solutions in the more time-efficient way than manual method.

4. Result and Discussion

4.1. Result of VRPTW

Based on the methodology and the assistance of C⁺⁺, the 7-route-network is discovered.

The table describes that the total travelled distance of 7 routes is about 500 Km. All the routes have more than one point, they are called "multiple routes". Route 4 is the longest with the length of 86.6 Km. Both route 2 and 7 have the largest numbers of points (running across 05 positions). Because the connections between points are quite short, the total time to finish the transport mission are not the highest (9.3 hours and 9.0 hours respectively).

Table 2 Operating parameters of 7 proposed routes

Route	Total distance (km)	Total time to complete transport missions (hours)	Total customers' demands (packages)
1. K-3-1-K	70,3	12,65	14
2. K-7-4-13-11-14-K	67,1	9,23	14
3. K-6-9-K	46,9	6,91	10
4. K-24-16-21-10-K	86,6	10,46	14
5. K-5-18-8-K	69,1	9,52	12
6. K-2-23-19-22-K	83,0	10,37	11
7. K-25-15-20-12-17-K	80,3	8,99	13
Sum	503,3	68,15	88

Note: K is abbreviation of Depot

The total time to finish 7 transport missions are 68.15 hours included the waiting time, unloading time at customers' points and time to travel between locations. In comparison with other routes, the route 1 has the largest time to finish transport mission (12.65 hours) although it has only

two points and the length of 70.3 Km. Reason is the high level of waiting time.

When it comes to the number of vehicles, because there are 7 routes, 7 trucks are enough.

4.2. Compare between current practice and result

There are several different main points between two solutions as follows:

- *The total travelled distance:* As a result of more optimal connection between points, the result has the smaller distance with 40.4 Km (about 8%) than the present solution's one. This allows saving operating cost related to tyre, fuel, maintenance and workers/drivers.

- *Load factor in terms of volume:* The factor of the result is larger at about 16%. This means capacity of truck in the result is used more effectively.

- *The number of routes and vehicles:* The route number of the result is one as large as the one of the current practice because all the routes in the result are multiple with at least two points while there is one single route in the present network. In fact, during a day, it is very difficult to dispatch one vehicle for two routes simultaneously, so the number of routes is equal the number of vehicles. The network of ICD Tien Son needs more vehicle than proposed solutions based on VRPTW.

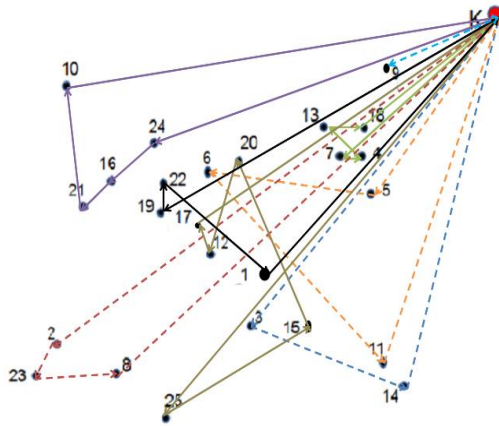
Table 3 Operating parameters of the result and current practice

	Parameters on the routes	Value		Difference $\Delta = (1) - (2)$
		Present solution (1)	Result (2)	
1	Total travelled distance (Km)	543,7	503,3	40,4
2	Total waiting time (Hour)	38,823	39,808	-0,985
3	Total unloading time (Hour)	14,75	14,75	0
4	Total travelled time (Hour)	14,307	13,592	0,715
5	Total time to solve transport mission (Hour)	67,88	68,15	-0,27
6	Load factor in terms of volume	0,733	0,849	-0,116
7	The number of route (Route)	08	07	01
8	The number of single route (Route)	01	0	01
9	The number of vehicle (Truck)	08	07	01

- *The total time of travelled distance:* This time is the advantage of the proposed solution because it has the smaller total travelled distance.

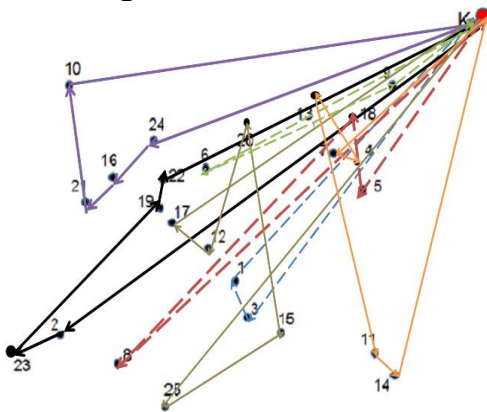
- *The total waiting time:* In comparison to

the result of VRPTW, the temporary routes are more advantageous in terms of waiting time. In multiple routes, vehicle usually waits until the accepted unloading time at points while in single one, there is no waiting time because vehicle positively chooses the starting time to reach the targeted point on time. Figure for waiting time of the result is approximately one hour (2.5%) higher than the one for present network.



Note: Dash line describes routes with less than 4 points

Fig. 4 Routes of the result



Note: Dash line describes routes with less than 4 points

Fig. 5 Current routes of ICD Tien Son

- *The total time to complete transport mission:* It is clear that the result has an advantage on the travelled time while the present network shows more benefit of waiting time. However, interestingly, the total time to solve transport missions of both solutions is nearly the same with the minor difference (0.5%).

The comprehensive assessment

Based on the comparison between indicated-above operating parameters, the better is the proposed network through application of VRPTW. The reason is that both two solutions complete transport mission in the nearly same time however the VRPTW result requires less input (total travelled distance, the number of trucks, the

number of routes).

5. Conclusion

In this paper, authors applied the heuristic procedure to solve VRPTW of ICD Tien Son. Starting with the classical saving algorithm, along with adding capacity and time window restraints at each customer point, a new solution for ICD Tien Son was found.

Theoretically, there is no evidence to affirm that the results of VRPTW are the optimal. Saving algorithm only concentrates on minimizing the total travelled distance, therefore the number of routes tends to decrease through constructing multiple ones. This possibly leads to an increase in waiting time and even the time to complete transport missions. In case of the time being so large, it is difficult to find the better network. Higher cost (for drivers, parking...) due to time increase may exceed the saved money thanks to decrease in travelled distance.

In case of ICD Tien Son, fortunately, both current solution and the result have the nearly similar time to complete transport mission, hence it is easier to find the better choice which needs less vehicles with smaller travelled kilometers. By contrast, it is necessary to calculate all the operating cost to make comparison and choice decisions. This is also the limitation of this study. As well as this, the method to calculate the distance between points is also a shortcoming. Many of distances in the research were inherited from the records of the company without being re-examined. The remainder were estimated through Google Maps. However, the problem is that authors mainly used added points on the map by someone instead of finding the exact positions by ourselves.

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